

Elimination of Cervical Cancer

1. Introduction

- Cervical cancer (CC) is the 4th most common cancer in women worldwide, with incidence ranging from <5 to >35 cases per 100,000 women (Fig.1).¹
- Persistent infection with high-risk human papillomavirus (HPV), particularly types 16 and 18, causes >99% of cervical cancers.^{2,3}
- The World Health Organisation (WHO) aims to eliminate CC (<4 cases per 100,000 women) through the 90–70–90 strategy targeting HPV vaccination, screening and treatment (Fig.2).^{2,4}
- CC develops through detectable precancerous stages, enabling prevention and early detection (Fig.3).⁴

3. Primary Prevention

HPV Vaccination

- HPV vaccines prevent infection with high-risk HPV types (HPV16 and HPV18), responsible for ~70% of CC cases.^{2,3}
- WHO recommends vaccination of girls aged 9–14 before sexual debut.²
- A two-dose schedule is recommended for girls aged 9 to 14 years (0 and 6–12 months).
- Individuals aged 15 years and older require a three-dose schedule (0, 2, 6 months).
- High HPV vaccination coverage reduces HPV-16/18 infections (~90%) and high-grade cervical abnormalities (~85%), with herd protection in unvaccinated populations.^{3,5}
- Most vaccine impact data come from high-income countries, limiting applicability to regions with the highest CC burden.⁶

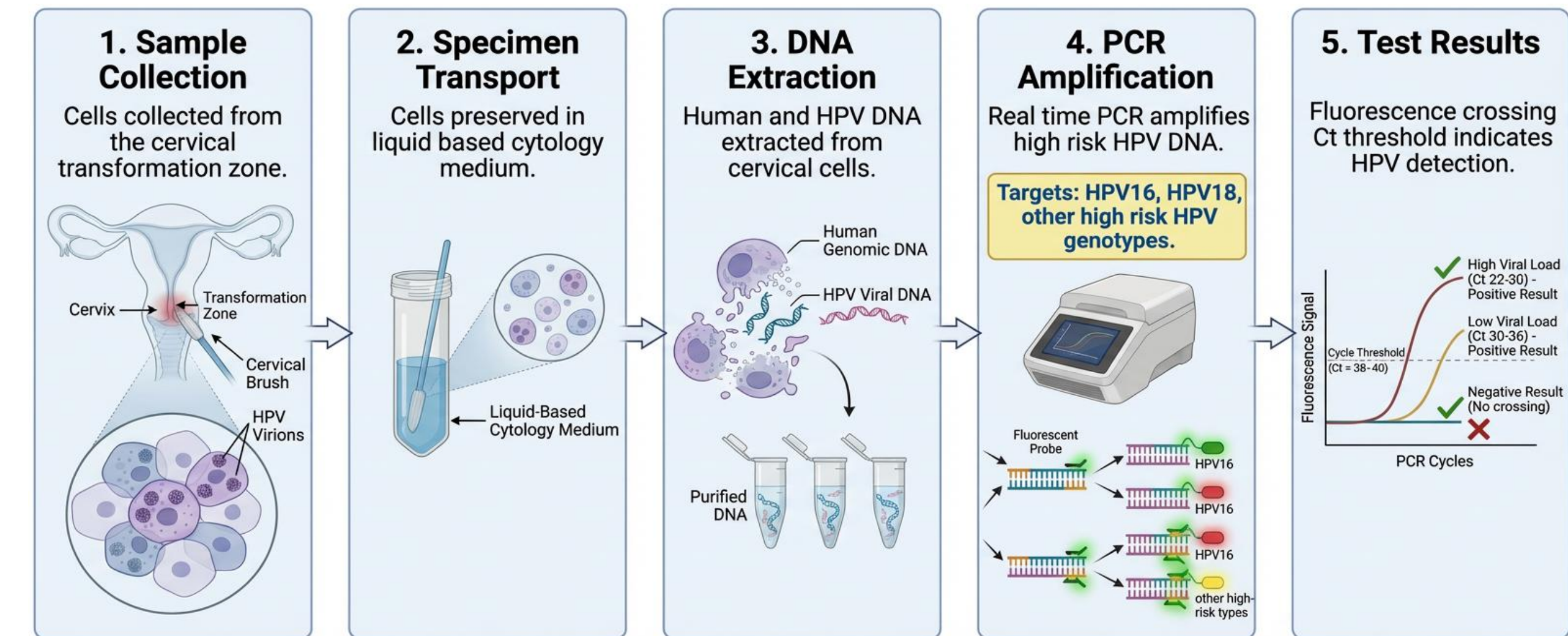


Figure 4. Workflow of HPV-based cervical screening showing sample collection, DNA extraction, PCR amplification, and detection of high-risk HPV genotypes. Adapted from ^{1,2}. Created in BioRender.com.

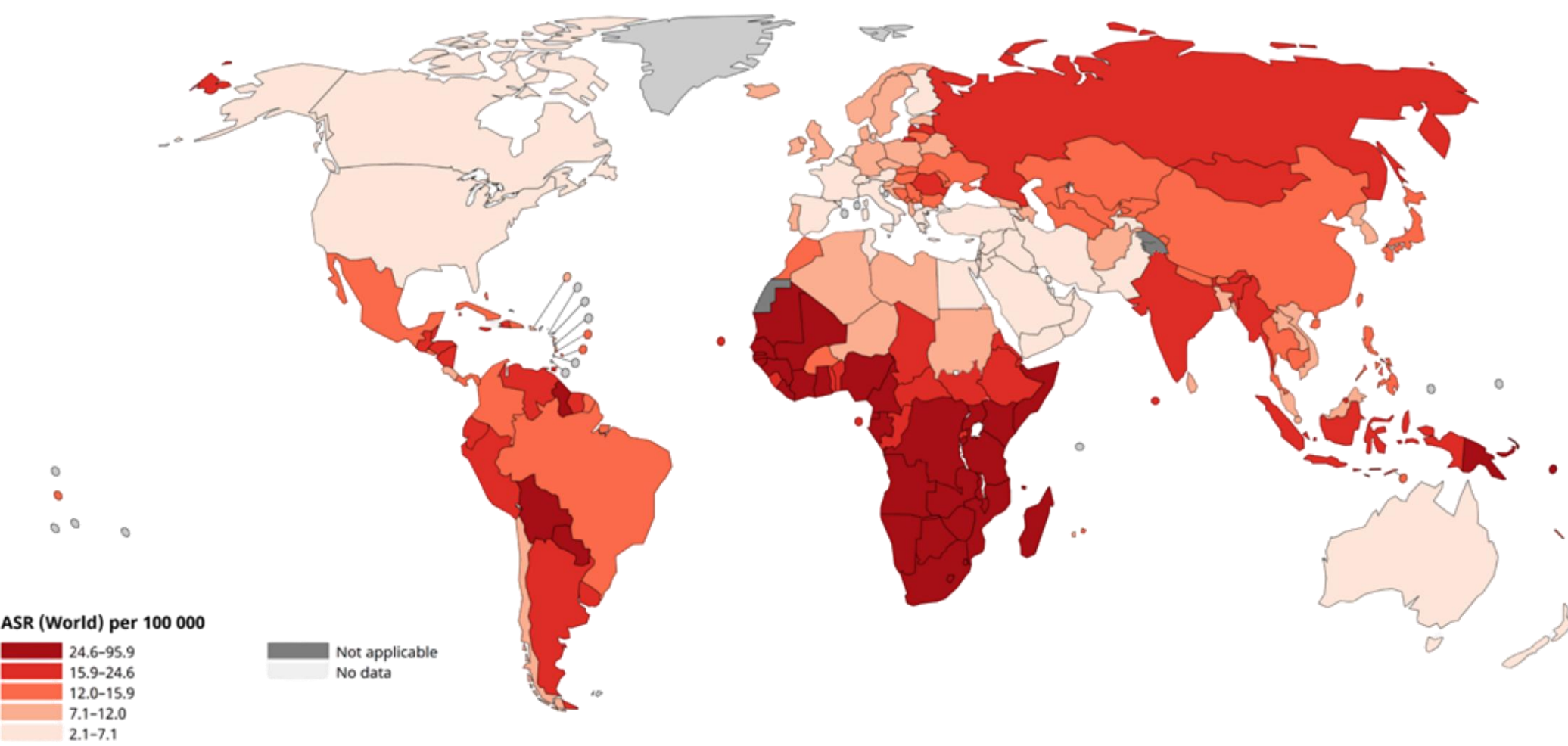


Figure 1. Global distribution of CC incidence in females, 2022, age-standardised rate per 100,000. Adapted from ². Created with BioRender.com.

2. Risk Factors³

- Persistent high-risk HPV infection drives cervical carcinogenesis.
- Five-year survival exceeds 90% for stage I disease but falls below 50% for stage IV, highlighting the importance of early detection.
- Smoking reduces HPV clearance and increases the risk of cervical intraepithelial neoplasia
- Multiple sexual partners or early sexual debut increase HPV acquisition
- Long-term oral contraceptive use modestly increases risk in women with persistent HPV.
- Immunosuppression, such as human immunodeficiency virus (HIV), increases HPV persistence and accelerates progression.

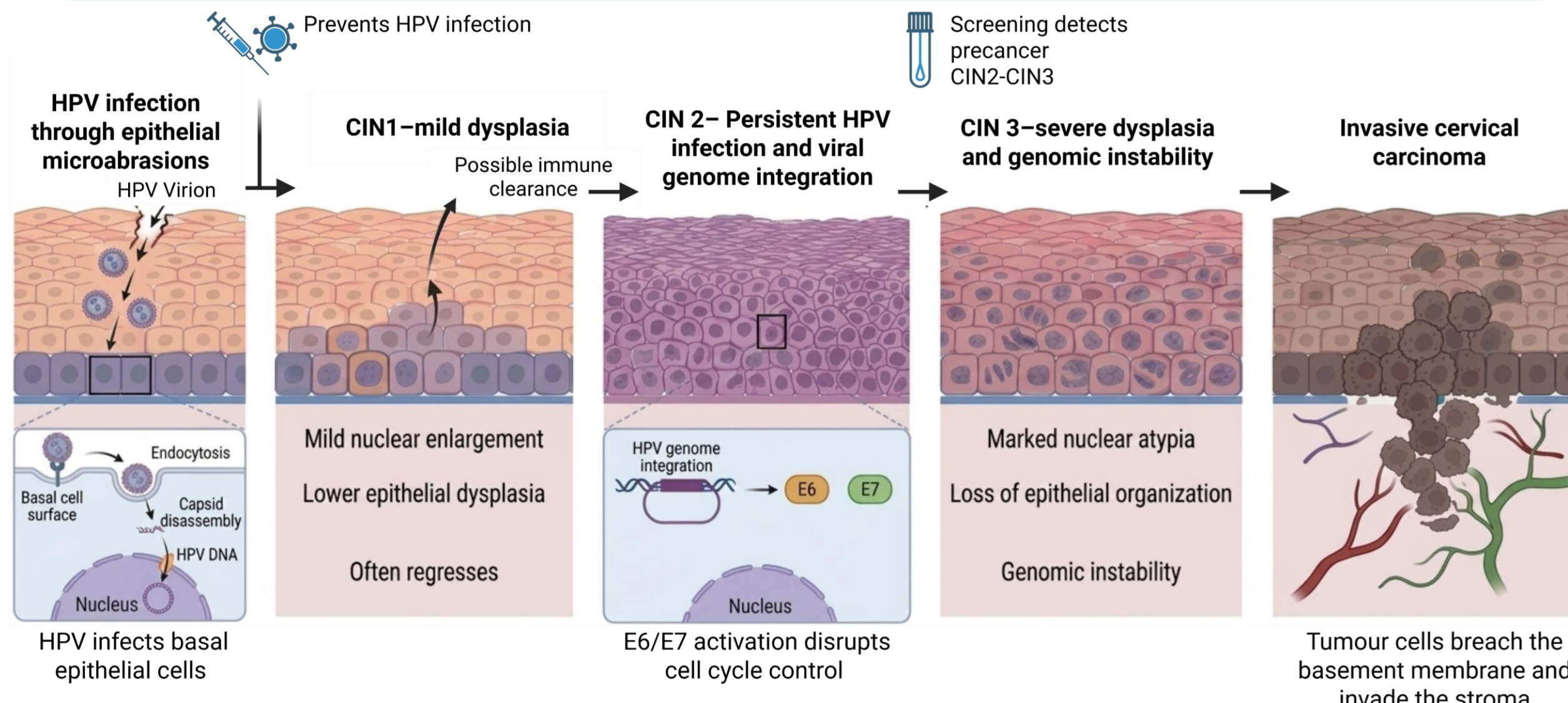


Figure 3. HPV-driven cervical carcinogenesis from basal epithelial infection to invasive carcinoma. Persistent infection with viral genome integration and E6/E7 activation leads to progression of cervical intraepithelial neoplasia (CIN) and basement membrane invasion. Adapted from ^{3,4}. Created in BioRender.com

4. Secondary Prevention

Cervical Screening

- Screening identifies CIN lesions before invasive cancer develops.^{4,7}
- Cytology sensitivity for CIN2+ is only ~50–60%, which historically limited early detection compared with HPV-DNA testing.⁷
- HPV testing has high sensitivity (~95%) but lower specificity (~85–90%), leading to more follow-up colposcopies.⁷
- HPV DNA testing provides 60–70% greater protection against invasive CC than cytology and allows screening intervals ≥5 years after a negative test.^{7,8}

5. WHO Screening Recommendations⁷

- HPV DNA testing is the preferred primary screening method (Fig.4).
- Screening should begin at age 30 years.
- Women aged 30–49 years should be prioritised for CC screening.
- When HPV testing is used, the recommended screening interval is 5–10 years.
- Women with a positive HPV test require triage, using HPV16/18 genotyping, cytology, visual inspection with acetic acid (VIA), or colposcopy (Fig.5).

6. Screening Programme Evaluation

- Screening estimates may be inflated by lead time, length and self-selection bias, which can overestimate survival benefits without reducing mortality.⁹
- Global HPV vaccination coverage remains low (~15%), particularly in low and middle income countries (LMICs).¹⁰
- Screening coverage is ~40% globally compared with >90% in high income countries (HICs) and ~18.5% in LMICs.¹⁰
- Some regions still rely on VIA, associated with ~20% mortality reduction, compared with ~50% reduction with HPV testing.⁸
- Immunocompromised populations (e.g. HIV) have higher CC risk.⁷

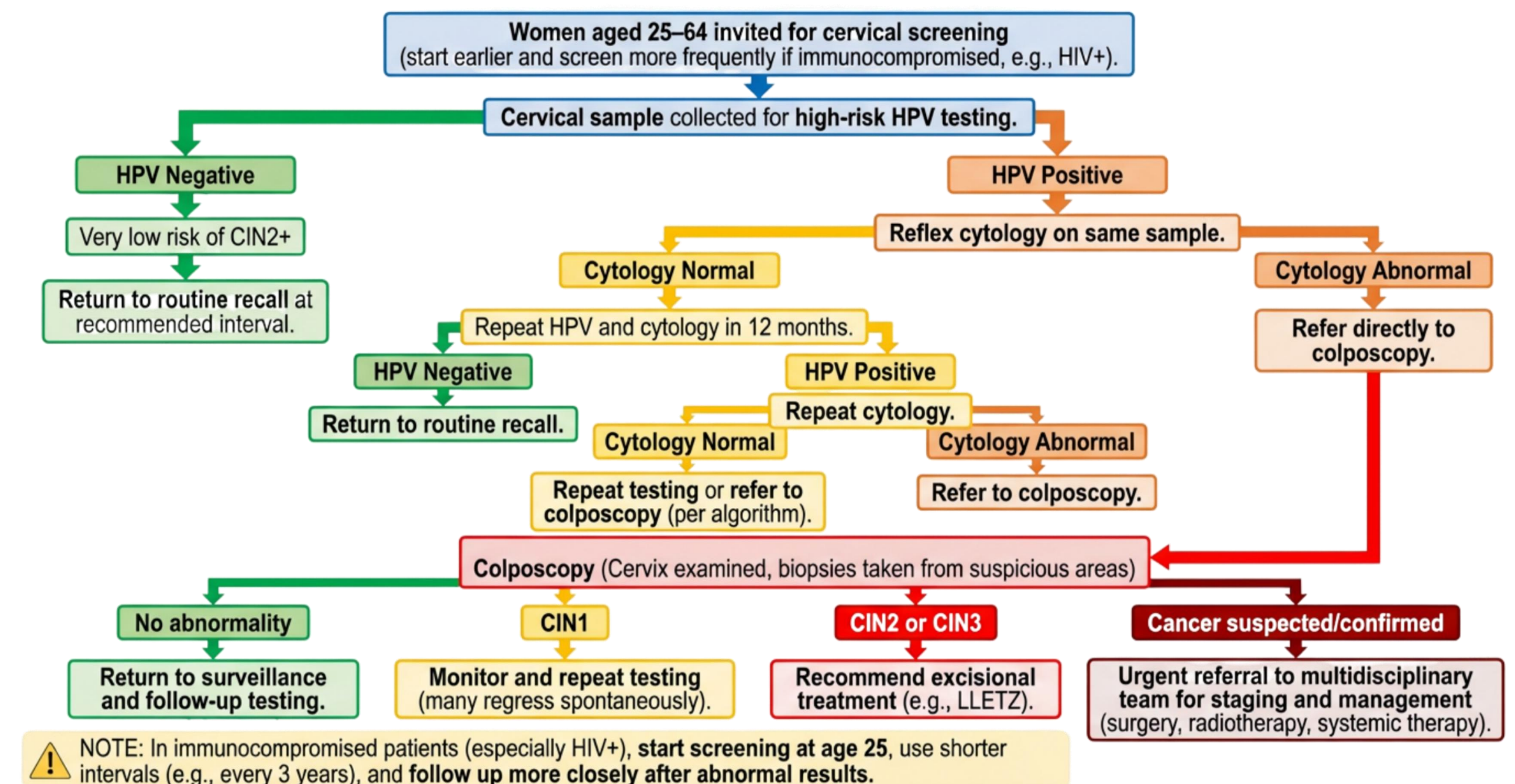


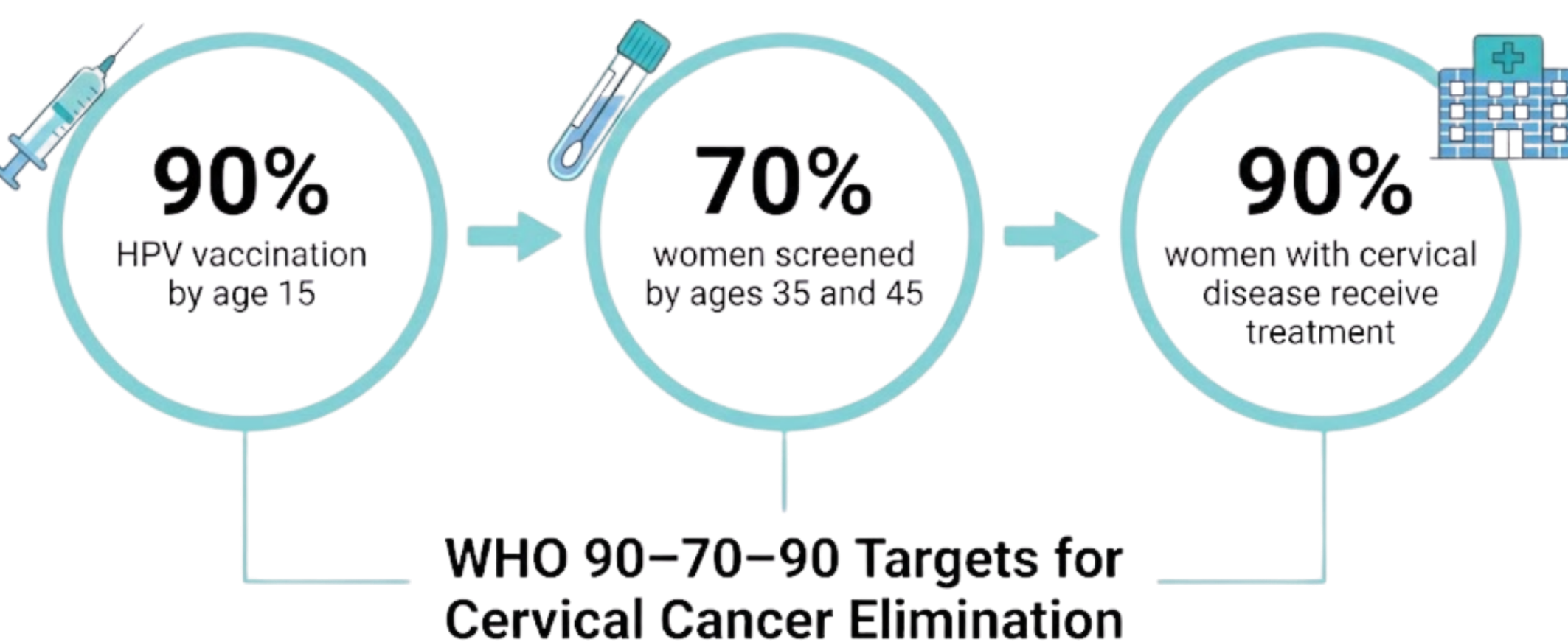
Figure 5. Cervical screening pathway in the UK showing HPV testing, cytology triage, colposcopy referral, and management of CIN lesions. Adapted from ⁴. Created in BioRender.com

7. Challenges to Cervical Cancer Elimination¹⁰

- Screening programmes are limited by infrastructure, funding, and trained personnel.
- Limited access to cancer treatment services, particularly radiotherapy and surgical treatment in LMICs.
- Socioeconomic inequalities, cultural barriers, and vaccine hesitancy reduce participation in vaccination and screening.

8. Future Directions and Conclusion

- CC is largely preventable through effective HPV vaccination and screening strategies.^{2,7}
- HPV vaccination and HPV-based screening enable early detection and treatment of precancerous lesions, reducing incidence and mortality.^{2,7}
- However, vaccination coverage and screening participation remain suboptimal in some regions.¹⁰
- Invitation-based interventions significantly increase screening attendance, with standard letters improving participation by ~71%.^{7,9}
- Personalised invitations, reminder systems, community outreach and HPV self-sampling can further improve screening uptake.⁹
- Future efforts should focus on expanding HPV vaccination, increasing screening participation, and improving access to care to achieve the WHO 90–70–90 elimination targets.¹⁰



WHO 90–70–90 Targets for Cervical Cancer Elimination

Figure 2. WHO CC elimination strategy (90–70–90 targets): 90% HPV vaccination by age 15, 70% screening coverage with high-performance tests, and 90% treatment of women with cervical disease. Adapted from ^{2,4,5}. Created in BioRender.com.